The SI unit and dimensions of T are $N\,m^{-1}$ and $M\,T^{-2}$, respectively.

Relation between surface tension and surface energy:

Consider a rectangular frame of wire ABCD in a soap solution (Figure 7.25). Let AB be the movable wire. Suppose the frame is dipped in soap solution, soap film is formed which pulls the wire AB inward due to surface tension. Let F be the force due to surface tension, then

$$F = (2\mathrm{T})l$$

here, 2 is introduced because it has two free surfaces. Suppose AB is moved by a small distance Δx to new a position $A'B'$. Since the area increases, some work has to be done against the inward force due to surface tension.

Work done = Force × distance = $(2\mathrm{T}\,l)\,(\Delta x)$

Increase in area of the film

$$\Delta A = (2l)\,(\Delta x) = 2l\,\Delta x$$

Therefore,

$$\text{surface energy} = \frac{\text{work done}}{\text{increase in surface area}}$$

$$= \frac{2Tl\,\Delta x}{2l\,\Delta x} = T \qquad (7.28)$$

Hence, the surface energy per unit area of a surface is numerically equal to the surface tension.

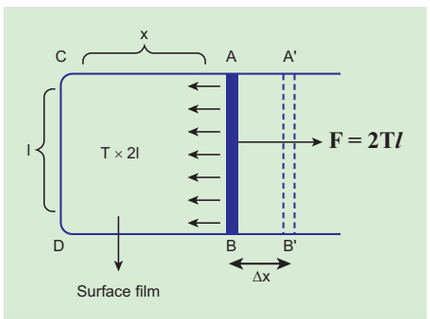


Figure 7.25 A horizontal soap film on a rectangular frame of wire ABCD

Note

It should be remembered that a liquid drop has only one free surface. Therefore, the surface area of a spherical drop of radius r is equal to $4\pi r^2$, whereas, a bubble has two free surfaces and hence the surface area of a spherical bubble is equal to $2 \times 4\pi r^2$.

EXAMPLE 7.10

Let $2.4\times10^{-4}\,J$ of work is done to increase the area of a film of soap bubble from 50 cm^2 to 100 cm^2. Calculate the value of surface tension of soap solution.

Solution:

A soap bubble has two free surfaces, therefore increase in surface area $\Delta A = A_2 - A_1 = 2(100-50) \times 10^{-4} m^2 = 100 \times 10^{-4} m^2$.

Since, work done $W = T \times \Delta A \Rightarrow T = \dfrac{W}{\Delta A}$

$$= \frac{2.4\times10^{-4}\,\mathrm{J}}{100\times10^{-4}\,\mathrm{m}^2} = 2.4\times10^{-2}\,\mathrm{N\,m^{-1}}$$

7.5.4 Angle of contact

When the free surface of a liquid comes in contact with a solid, then the surface of the liquid becomes curved at the point of contact. Whenever the liquid surface becomes a curve, then the angle between the two medium (solid-liquid interface) comes in the picture. For an example, when a glass plate is dipped in water with sides vertical as shown in figure, we can observe that the water is drawn up to the plate. In the same manner, instead of water the glass plate is dipped in mercury, the surface is curved but now the curve is depressed as shown in Figure 7.29

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